FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – IV
PAPER –2
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. C

Sol. For upper portion:

$$f_1 = \frac{\frac{\mu_2}{\mu_1} - 1}{\frac{\mu_2}{\mu_1} - 1}(f_a) = \frac{1.5 - 1}{\frac{1.5}{1.2} - 1} \times 20 = 40 \text{ cm}$$

For lower portion, $f_1 = \frac{\frac{\mu_2}{\mu_3} - 1}{\frac{\mu_2}{\mu_3} - 1}(f_a)$

$$= \frac{1.5 - 1}{\frac{1.5}{2.5} - 1} \times 20 = -25 \text{ cm}$$

$\because$ Object is at infinity, images will be form at focal points.

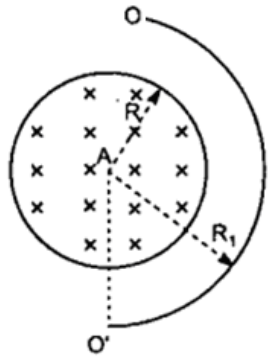
Hence the distance between two images will = $40 + |-25| = 65$ cm

2. B

Sol. $E = \frac{R^2}{2R_1}\left(\frac{dB}{dt}\right)$

$$\text{e.m.f} = \frac{\theta}{2\pi}\frac{R^2}{2R_1}\left(\frac{dB}{dt}\right)2\pi R_1$$

$$= \frac{\theta}{2}R^2\left(\frac{dB}{dt}\right)$$



3. D

Sol. As $\gamma > 1$

for $TV^{\gamma-1} = \text{constant}$ $\frac{dT}{dv} < 0$

and for $T^{\gamma} = KP^{\gamma-1}$

$\frac{dT}{dP} > 0$

4. A

Sol. $y = \frac{0.8}{3(x+2t)^2+4}$

v = 2 m/s

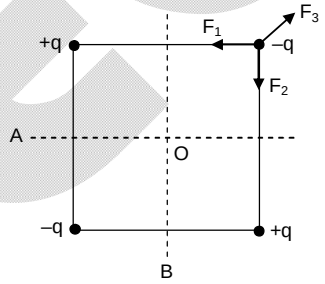
A = 0.2 m

5. AC

Sol. AOB is zero potential surface

Induce charge = –q

$F = \frac{kq^2}{a^2}\sqrt{2} - \frac{kq^2}{2a^2}$



6. AC

Sol. $I_1\left[\frac{n(n+1)}{2}r\right] + Ix = nE$ where x external resistance

$I_m\left(\frac{n(n+1)}{2}r\right) + I_x = mnE$

$I_1 + I_2 + I_3 + \ldots\ldots + I_m = I$

$\Rightarrow I = \frac{\frac{n(m+1)E}{2}}{x + \frac{n(n+1)}{2m}r} = \frac{V}{x+R}$

7. CD

Sol. $\frac{dQ}{dt} = \frac{KA(\Delta T)}{x} = \frac{KA[0-(-\theta)]}{x}$; $\frac{dQ}{dt} = \frac{KA\theta}{x}$

8. ACD

Sol. $y(x, t) = -\cos kx \sin \omega t$

$y(0.05, 0.05) = -4\cos\left(\frac{2\pi}{0.4}0.05\right)\sin\left(\frac{2\pi}{0.2}0.005\right)$

$= -4\cos\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{2}\right) = -2\sqrt{2}$ cm

$v = f\lambda = \frac{0.4}{0.2}$ = 2 m/s

$\frac{\partial y}{\partial t} = -A\,\omega\cos(kx)\cos(\omega t)$

$= -4\left(\frac{2\pi}{0.2}\right)\cos\left(\frac{2\pi}{0.4}\frac{1}{15}\right)\cos\left(\frac{2\pi}{0.2}0.1\right)$

$\frac{\partial y}{\delta t}$ = 20 π cm/sec = particle velocity

9. AC

Sol. In SHM, time taken to move from x = 0 to $x = \frac{A}{2}$ is $\frac{T}{12}$ and time taken to move from extreme to $x = \frac{A}{2}$ is $\frac{T}{6}$.

If $\frac{T}{12}$ = 1 $\Rightarrow$ T = 12

If $\frac{T}{6}$ = 1 $\Rightarrow$ T = 6.

10. ABC

Sol. $E_s = \frac{-GMm}{R}$; $E_1 = \frac{-GMm}{2(2R)}$; $E_2 = \frac{-GMm}{2(3R)}$

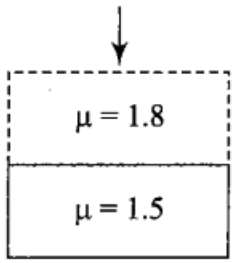
$E_1 - E_s = \frac{3}{4}mgR$

$E_2 - E_1 = \frac{mgR}{12}$

Section – B

11. 9

Sol. Path difference between rays reflected from upper and lower faces of layer = 2μt cosr = 2 μt (for normal incidence). But there is change in path of $\lambda/2$ of light at upper surface due to reflection from denser medium. So actual path difference is 2 μt – $\lambda/2$.



For constructive interference $2\mu t - \frac{\lambda}{2} = n\lambda$

$t = \frac{(2n+1)\lambda}{4\mu}$. For least thickness n = 0.

$\therefore \quad t_{min} = \dfrac{\lambda}{4\mu} = \dfrac{648}{4\times 1.8}\text{nm} = 90 \text{ nm}$

12. 5

Sol. $E_n = -\dfrac{mZ^2e^4}{8\varepsilon_0^2 n^2h^2}$, so $hf = +\dfrac{mZ^2e^4}{8\varepsilon_0^2h^2}\left[\dfrac{1}{16}-\dfrac{1}{25}\right]$

$\therefore \quad f = \dfrac{mZ^2e^4}{8\varepsilon_0^2h^3}\left[\dfrac{9}{16\times 25}\right]$

and frequency $f_4 = \dfrac{Z^2e^2m}{4\varepsilon_0^2n^3h^3} = \dfrac{Z^2e^4m}{4\varepsilon_0^2(4)^3h^3}$

$\therefore \quad \dfrac{f}{f_4} = \dfrac{18}{25}$, so m = 5.

13. 4

Sol. When the two spheres come in contact, they exert equal and opposite impulse.

$$\int Fa\, dt = I_1(\omega_0 - \omega_1) \qquad \ldots(i)$$

$$\int Fb\, dt = I_2\omega_2 \qquad \ldots(ii)$$

Finally $\omega_1 a = \omega_2 b$ …(iii)

$$I_1(\omega_0 - \omega_1) = I_2\frac{a^2}{b^2}\omega_1$$

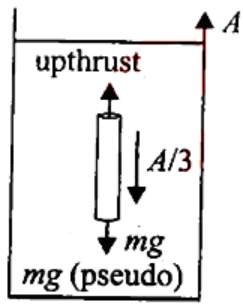
$$\omega_1 = \frac{I_1\omega_0}{I_1 + \dfrac{a^2}{b^2}I_2} = 4 \text{ rad/s.}$$

14. 5

Sol. Let the volume of the cylinder be V. When the cylinder is floating , upthrust = weight. Hence,

$$\rho\left(\frac{3}{4}V\right)g = mg \Rightarrow V = \frac{4m}{3\rho}$$

Let the acceleration of the particle vessel be A (upwards). In the reference frame of the vessel, the acceleration of the cylinder is A/3.



$\therefore$ mg + mA – upthrust = $m\left(\dfrac{A}{3}\right)$

mg + mA – ρVg′ = $m\left(\dfrac{A}{3}\right)$

where g′ = g + A = effective value of g for upthrust.

$\therefore$ mg + mA – ρV (g + A) = $m\left(\dfrac{A}{3}\right)$

$\Rightarrow \quad mg - \dfrac{4}{3}m(g+A) = -m\dfrac{2}{3}A$

$A = \left(-\frac{g}{2}\right)$ upwards

The acceleration of the vessel should be $\frac{g}{2} = \frac{10}{2} = 5\ ms^{-2}$ (downwards).

15. 6

Sol. Let the pressures in wide and narrow limbs be P_1 and P_2, respectively. If R_1 and R_2 be the radii of meniscus in wide and narrow limb, pressure just below the meniscus of wide tube = $P_1 - \frac{2T}{R_1}$ and pressure just below the meniscus of narrow limb = $P_2 - \frac{2T}{R_2}$.

Therefore, difference of these pressures

$$\left(P_1 - \frac{2T}{R_1}\right) - \left(P_2 - \frac{2T}{R_2}\right) = h\rho g$$

Therefore, true pressure difference,

$$P_1 - P_2 = h\rho g - 2T\left(\frac{1}{R_2} - \frac{1}{R_1}\right)$$

For the water and glass surface, taking the angle of contact θ to be zero, we have $R_1 = \frac{r_1}{\cos\theta} = r_1$ and $R_2 = \frac{r_2}{\cos\theta} = r_2$ where r_1 and r_2 are radii of wide and narrow limbs, respectively.

$\therefore$

$$P_1 - P_2 = h\rho g - 2T\left(\frac{1}{r_2} - \frac{1}{r_1}\right) = 0.2\times10^3\times9.8 - 2\times72\times10^{-3}\times\left(\frac{1}{7.2\times10^{-4}} - \frac{1}{1.44\times10^{-3}}\right)$$

$= 1.96 \times 10^3 - 0.10 \times 10^3 = 1.86 \times 10^3\ N/m^2 = 1860\ N/m^2$

$\therefore$ N = 6.

16. 8

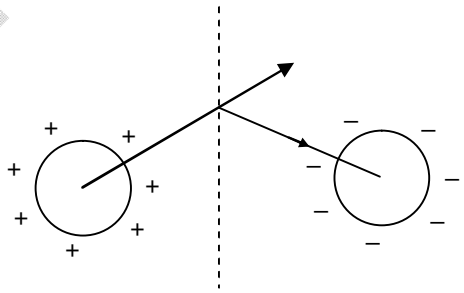
Sol. $J = \sigma E = \frac{\sigma q \ell}{2\pi\varepsilon_0\varepsilon r^3}$

$$I = \int_0^\infty J\,2\pi x dx = \frac{q}{\varepsilon_0\varepsilon\rho}$$

$$\Delta V = \frac{q}{2\pi\varepsilon_0\varepsilon a} = IR$$

$\Rightarrow \quad R = \frac{\rho}{2\pi a} = 0.50$

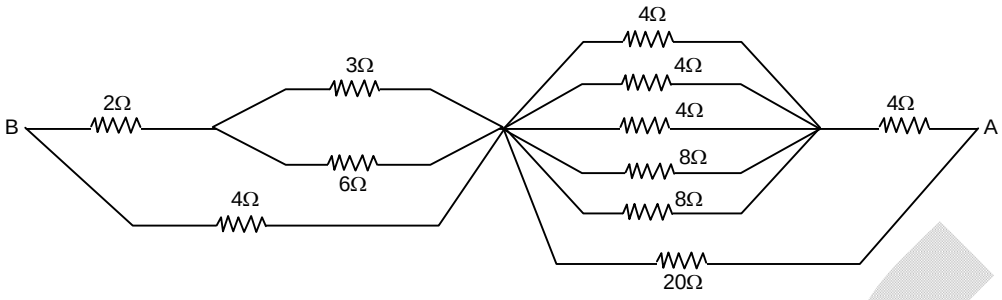
$\therefore$ n = 8.



17. 4

Sol. $C_{eq} = \frac{2C_1C_2 + C_2C_3 + C_3C_1}{C_1 + C_2 + 2C_3} = \frac{7}{5}$

Where $C_1 = 1$, $C_2 = 2$, $C_3 = 1$



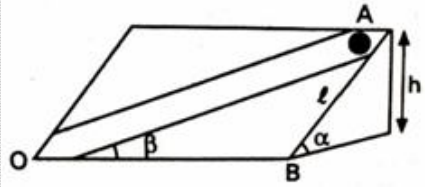
$R_{eq} = 6$

$\tau = RC = 8.4$

$\therefore$ n = 4.

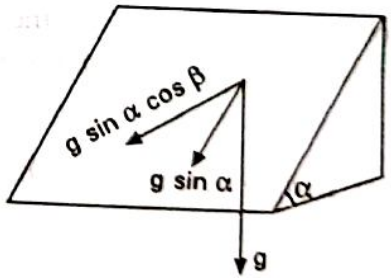
18. 2

Sol. Component of acceleration along incline is g sin α and effective acceleration along groove is a = sin α cos β.



From figure $\ell = \dfrac{h}{\sin\alpha} = OB$

OA is the groove making an angle β = 45° with AB on the inclined surface.



From figure, $\dfrac{OB}{OA} = \cos\beta$

or $OA = \dfrac{h}{\sin\alpha\cos\beta}$

Since acceleration along groove is constant, we may use equation $s = \dfrac{1}{2}at^2$.

$\therefore$ k = 2.

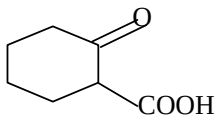
Chemistry

PART – II

Section – A

19. C

Sol.

Intermediate is 

Which loses CO_2 on heating (β-keto acid)

20. B

Sol. M absorbs some heat energy and prevents decomposition of O_3

21. C

Sol. $T_f = 4T_i$

$$\Delta S = nC_V \ln\frac{T_f}{T_i} + nR\ln\frac{v_f}{v_1}$$

$$0 = n\times\frac{3R}{2}\ln 4 + n.R\ln\frac{v_f}{v_1}$$

$$\frac{3}{2}\log 4 = \log\frac{v_i}{v_f}$$

$$4^{3/2} = \frac{v_i}{v_f}$$

$$8v_f = v_i$$

$$0 = \Delta S = nC_p \ln\frac{T_f}{T_i} + nR.\ln\frac{Pi}{P_f}$$

$$4^{5/2} = \frac{P_f}{P_i}$$

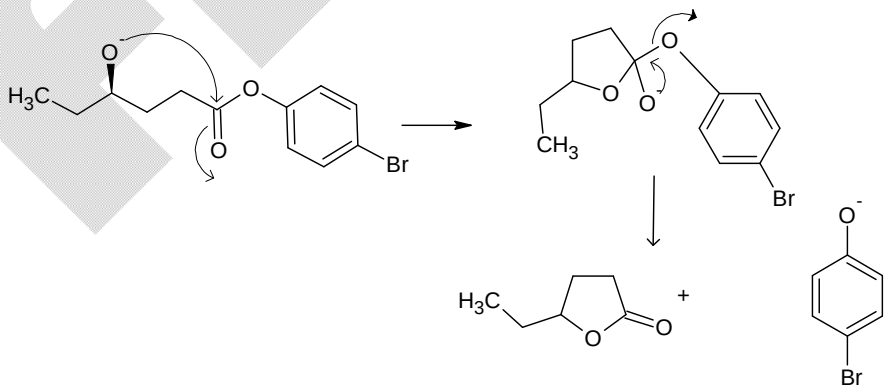
$$32P_i = P_f$$

22. D

Sol. At ¼ th and 3/4 th neutralization points the solution is buffer solution. Upon substitution we can get the ans.

23. ACD

Sol.



24. AC

Sol. Azeotropic mixture has fixed composition.

25. ABC

Sol. Rate = $k[X][Y]^0 \Rightarrow 2 \times 10^{-8} = k[0.1]\,[0.1]^0 \Rightarrow k = 2 \times 10^{-7} s^{-1}$

26. ABC

Sol. $CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$

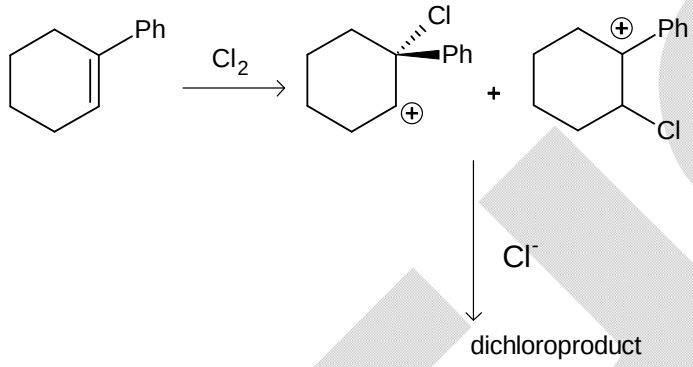
$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$

$Na_2S + 2H_2O \longrightarrow 2NaOH + H_2S$

$KNO_3 + H_2O \longrightarrow$ No reaction

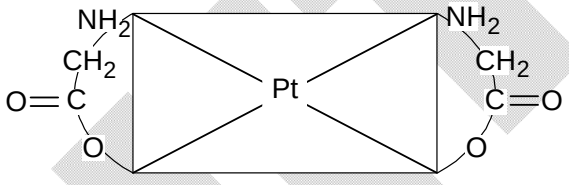
27. ABCD

Sol.

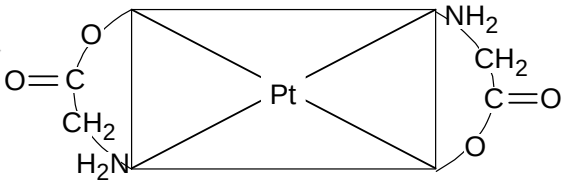


28. BCD

Sol. The geometrical isomers of (B) are

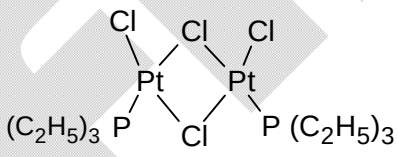


Cis

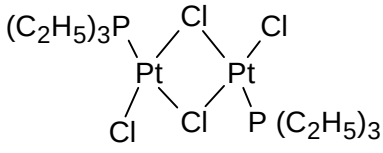


trans

The geometrical isomers of (C) are

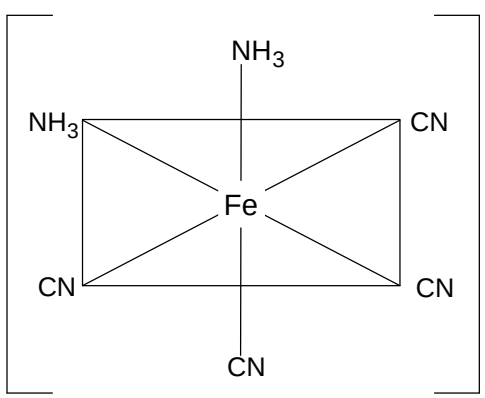


Cis

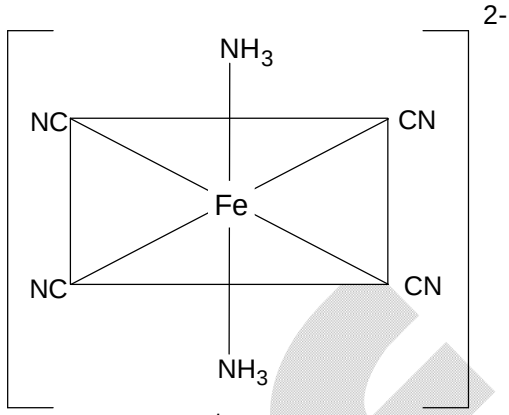


trans

The geometrical isomers of (D) are



Cis

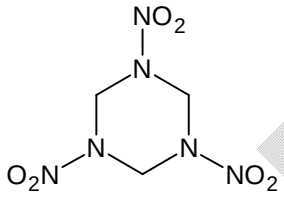


trans

Section – B

29. 6

Sol. Compound (Z) is



30. 4

Sol. $(4n + 2)\pi e^-$ rule

31. 3

Sol. $\eta^5 C_2H_5 - 5e^-$
2 CO – $4e^-$
Mo – $6e^-$
Mo – Mo – $3e^-$

32. 3

Sol. $S^{2-} + HCl \longrightarrow H_2S \xrightarrow[NaOH]{Na_2[Fe(CN)_5NO]} Na_4[Fe(CN)_5(NOS)]$

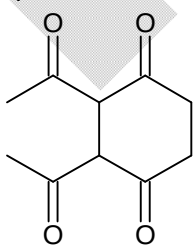
33. 5

Sol. $$\frac{25\times10^{-3}(g)}{250} = \frac{16\times10^{-3}(g)}{250 - x18}$$

X = 5

34. 4

Sol.



35. 4

Sol. (1) ala-phe-leu-gly
(2) ala-phe-gly-leu
(3) ala-leu-phe-gly
(4) ala-leu-gly-phe

36. 3

Sol. It exists in two geometrical isomeric forms in which trans isomer is optically active.

FIITJEE

Mathematics

PART – III

Section – A

37. A

Sol. Since for two square matrices A & B the trace of AB – BA is zero.

$\therefore$ let $AB - BA = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$

$\Rightarrow$ $(AB - BA)^2 = k\, I_2$

where $k = a^2 + bc$

and odd power of AB – BA is equal to a multiple of this matrix.

i.e. odd power it can not be equal to I_2

$\therefore$ 'n' is even.

38. B

Sol. $A^2B^2 = rB^4$

$B^2A^2 = rB^4$

$\Rightarrow A^2B^2 = B^2A^2$

& $rB^2A = rAB^2 = A^3$

Multiplying $pAB + qBA = I_n$ on the right and then on the let by B, we obtain

$pAB^2 + qBAB = B$

$pBAB + qB^2A = B$

also we have $B^2A = AB^2$

on subtraction $(p - q)(AB^2 - BAB) = 0$

if $p \neq q$ then $AB^2 = BAB$

$\therefore (p + q)\, AB^2 = (p + q)\, B^2A = B$

$\Rightarrow (p + q)\, A^2B^2 = AB$

& $(p + q)B^2A^2 = BA$

$\Rightarrow AB = BA$

which is a contradiction

$\therefore p = q$

39. A

Sol. Integrating the given differential equation, we have $\frac{dy}{dx} = \frac{-\cos 3x}{3} + e^x + \frac{x^3}{3} + C_1$ but

$y_1(0) = 1$ so $1 = \left(\frac{-1}{3}\right) + 1 + C_1 \Rightarrow C_1 = \frac{1}{3}$.

Again integrating, we get $y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x + C_2$

but $y(0) = 0$ so $0 = 0 + 1 + C_2 \Rightarrow C_2 = -1$. Thus $y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$

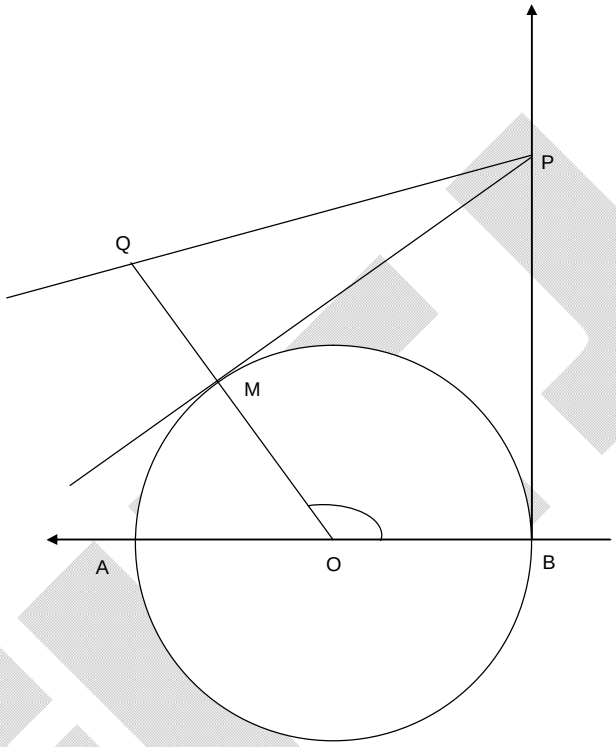
40. B

Sol. Let the radius of the circle is equal to 1. Set the origin at B with BA the positive x – semi axis and t the y – axis. If $\angle BOM = \theta$, then $BP = PM = \tan\frac{\theta}{2}$. In triangle PQM, PQ $= \frac{\tan\frac{\theta}{2}}{\sin\theta}$. So the coordinates of Q are $\left(\frac{\tan\frac{\theta}{2}}{\sin\theta}, \tan\frac{\theta}{2}\right) = \left(\frac{1}{1+\cos\theta}, \frac{\sin\theta}{1+\cos\theta}\right)$.

The x and y coordinates are related as follows:

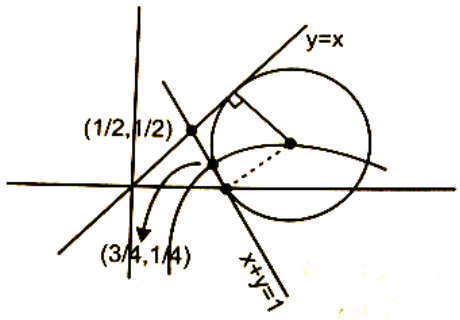
$$\left(\frac{\sin\theta}{1+\cos\theta}\right)^2 = \frac{1-\cos^2\theta}{(1+\cos\theta)^2} = \frac{1-\cos\theta}{1+\cos\theta} = 2\frac{1}{1+\cos\theta} - 1.$$

Hence the locus of Q is the parabola $y^2 = 2x - 1$.



41. BC

Sol. $4a = 2\left(\frac{|0-1|}{\sqrt{2}}\right) = \frac{2}{\sqrt{2}} = \sqrt{2}$



42. BCD

Sol. Put $(a - r\cos\theta, 2 - r\sin\theta)$ to equn. of ellipse

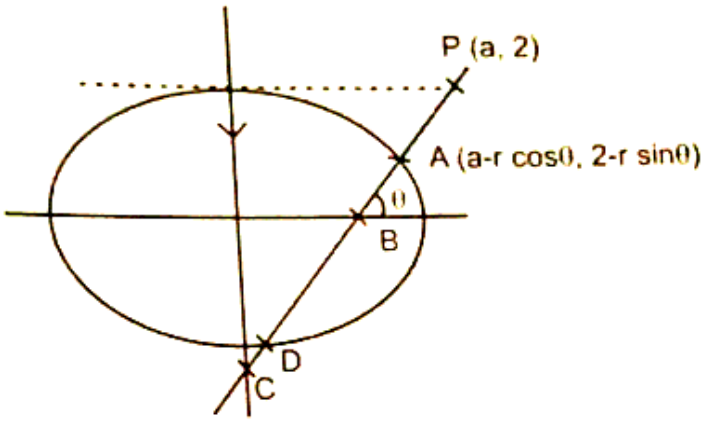
$$\Rightarrow \frac{a^2 + r^2\cos^2\theta - 2ar\cos\theta}{9}$$

$$+\frac{4 + r^2\sin^2\theta - 4r\sin\theta}{4} - 1 = 0$$

$$= \frac{4a^2}{4\cos^2\theta + 9\sin^2\theta}$$

Put $(a - r\cos\theta, 2 - r\sin\theta)$ to equation $xy = 0$

$$\Rightarrow (PA)(PD) = \frac{\frac{a^2}{9}}{\frac{\cos^2\theta}{9} + \frac{\sin^2\theta}{4}}$$



$$\Rightarrow 2a + r^2\sin\theta\cos\theta - (a\sin\theta + 2\cos\theta)r = 0$$

$$\Rightarrow (PB)(PC) = \frac{2a}{\sin\theta\cos\theta}$$

$PA.PD = PB.PC$

$$\Rightarrow \frac{4a^2}{4\cos^2\theta + 9\sin^2\theta} = \frac{2a}{\sin\theta.\cos\theta}$$

$$\Rightarrow a = \frac{4\cot\theta + 9\tan\theta}{2}$$

$$\Rightarrow |a| \geq \sqrt{4 + 9} = 6$$

$$\Rightarrow a \in (-\infty, -6] \cup [6, \infty)$$

43. CD

Sol. If α is a multiple of π, then $I(\alpha)=0$. Otherwise, use the substitution $x=\cos\alpha+t\sin\alpha$. The indefinite integral becomes

$$\int\frac{\sin\alpha\,dx}{1-2x\cos\alpha+x^2}=\int\frac{dt}{1+t^2}=\text{art tan}\,t+C.$$

$$\therefore I(\alpha)=\arctan\left(\frac{1-\cos\alpha}{\sin\alpha}\right)-\text{arc tan}\left(\frac{-1-\cos\alpha}{\sin\alpha}\right),$$

where the angles are to be taken between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$. But

$$\frac{1-\cos\alpha}{\sin\alpha}\times\frac{-1-\cos\alpha}{\sin\alpha}=-1.$$

Hence the difference between these angles is $\pm\frac{\pi}{2}$. Notice that the sign of the integral is the same as the sign of α. Hence $I(\alpha)=\frac{\pi}{2}$ if $\alpha\in(2k\pi,(2k+1)\pi)$ and $-\frac{\pi}{2}$ if $\alpha\in((2k+1)\pi,(2k+2)\pi)$ for some integer k.

44. AD

Sol. $k_1u-k_2v=0$ …….(i)

$k_1u+k_2v=0$ …….(ii)

$\because$ equations of bisectors of the angles formed by lines (i) and (ii) are

$$\frac{k_1u-k_2v}{\sqrt{(ak_1-bk_2)^2+(k_1b+ak_2)^2}}=\frac{\pm(k_1u+k_2v)}{\sqrt{(k_1a+bk_2)^2+(k_1b-ak_2)^2}}$$

$\Rightarrow k_1u-k_2v=\pm(k_1u+k_2v)$ …….(iii)

(i) by taking positive sign in (iii), we get

$k_1u-k_2v=k_1u+k_2v$

$2k_2v=0 \quad\Rightarrow\quad v=0$

(ii) by taking negative sign in (iii), we get

$u=0$

45. BD

Sol. After we bring the function into the form $f(x)=\dfrac{\left(x-1+\dfrac{1}{x}\right)^3}{x^3-1+\dfrac{1}{x^3}}$, let $x+\dfrac{1}{x}=s$

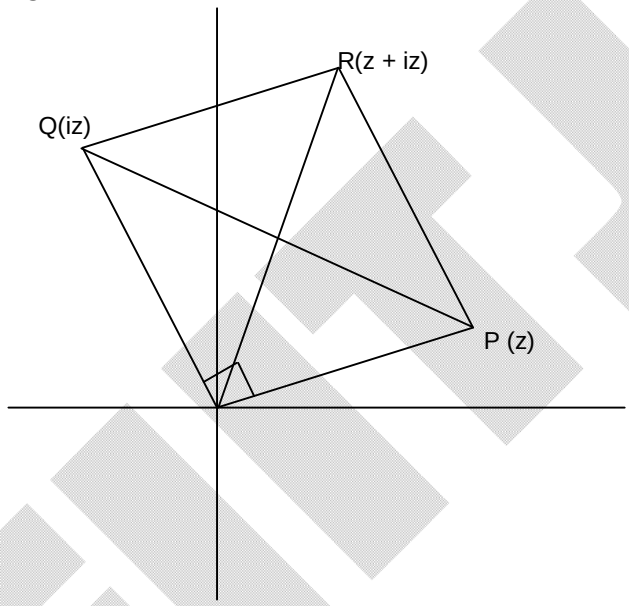
$h(s)=\dfrac{(s-1)^3}{s^3-3s-1}=1+\dfrac{-3s^2+6s}{s^3-3s-1}$ over the domain $(-\infty,-2]\cup[2,\infty)$. Setting the first derivative equal to zero yields the equation $3(s-1)(s^3-3s^2+2)=0$.

The roots are s = 1 (double root) and $s=1\pm\sqrt{3}$. Of these, only $s=1+\sqrt{3}$ lies in the domain of the function.
We compute

$\lim\limits_{x\to\pm\infty} h(s)=1,\ h(2)=1,\ h(-2)=9,\ h\left(1+\sqrt{3}\right)=\dfrac{\sqrt{3}}{2+\sqrt{3}}$.

46. AC

Sol.



Section – B

47. 3

Sol. Equation of the chord of contact of a point $P(3\sec\theta,\ 2\tan\theta)$ on the hyperbola with respect to the circle is $(3\sec\theta)x+(2\tan\theta)y=9$ (1)

Let M (h, k) be the mid point of (1), then equation of (1) in terms of the mid – point is

$hx+ky=h^2+k^2$ (2)

Since (1) and (2) represent the same line.

$\sec\theta=\dfrac{3h}{h^2+k^2}$, $\tan\theta=\dfrac{9k}{2(h^2+k^2)}$s

$\Rightarrow$ locus of (h, k) is

$\Rightarrow \dfrac{9x^2}{\left(x^2+y^2\right)^2}-\dfrac{81y^2}{4\left(x^2+y^2\right)^2}=1$

or $4\left(x^2+y^2\right)^2=bx^2-cy^2$

$\Rightarrow \quad a=4,\ b=36,\ c=81$

$a^2+b^2+c^2=16+1296+6561=7873$

48. 5

Sol. Let $r=\lambda b+\mu c$ and $c=\pm(xi+yj)$. Since c and b are perpendicular, we have

$4x+3y=0 \qquad \Rightarrow c=\pm x\left(i-\dfrac{4}{3}j\right)$

$\pm 1=$ proj. of r on $b=\dfrac{r.b}{|b|}=\dfrac{(\lambda b+\mu c).b}{|b|}=\dfrac{\lambda b.b}{|b|}$

$[\because b.c=0]$

$=\lambda|b|=5\lambda$. Hence $\lambda=\dfrac{1}{5}$

Also, $\pm 2=$ proj. of r on $c=\dfrac{r.c}{|c|}$

$=\dfrac{(\lambda b+\mu c).c}{|c|}=\mu|c|=\dfrac{5}{3}\mu x$

Thus, $\mu x=\pm\dfrac{6}{5}$. Therefore,

$r=\dfrac{1}{5}(4i+3j)+\dfrac{6}{5}\left(i-\dfrac{4}{3}j\right)=\pm(2i-j)$

$r=\dfrac{1}{5}(4i+3j)-\dfrac{6}{5}\left(i-\dfrac{4}{3}j\right)=\pm\left(-\dfrac{2}{3}i+\dfrac{11}{5}j\right)$

Thus there are four such vectors

$\sum_{i=1}^{4}|r_i|^2=2|2i-j|^2+2\left|-\dfrac{2}{5}i+\dfrac{11}{5}j\right|^2=20$

49. 1

Sol. Let $f(x)=\sin x+\tan x-2x$. Then $g(x)=f'(x)$

$=\cos x+\sec^2 x-2$

$\Rightarrow \quad g'(x)=-\sin x+2\sec^2 x\tan x$

$=\sin x\left(-1+\dfrac{2}{\cos^3 x}\right)$

$=\sin x\left(\dfrac{1-\cos^3 x+1}{\cos^3 x}\right)$

Since for $0 < x < \frac{\pi}{2}$, we have $0 < \cos^3 x < 1$, g is an increasing function. Hence $g(x) > g(0)$, i.e. $\cos x + \sec^2 x - 2 > 0$. Therefore f is an increasing function, so $f(x) > f(0)$ for $0 < x < \frac{\pi}{2}$. Hence $\sin x + \tan x > 2x$. Thus $g\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + 1$

50. 1

Sol. The inverse of 2 × 2 matrices $C = [C_{ij}]_{2\times 2}$ with integer entries is a matrix with integer entries if and only if $|c| = \pm 1$

$$\left\{C^{-1} = \frac{\text{adj}(C)}{|C|} \,\&\, \left|C^{-1}\right| = \frac{1}{|C|}\right\}$$

So lets take a polynomial.

$$P(x) = |A + xB|$$

as per the problem $P(0), P(1), P(2), P(3), P(4) \in \{-1, 1\}$

three of these must be same

and P(x) has degree at most 2.

$\therefore$ it is constt.

$\therefore |A + xB| = \pm 1, \ \forall x$

$\therefore |A + 5B| = \pm 1$

51. 2

Sol. $a_1 + a_3 + \ldots + a_{99} = 50$

$$\Rightarrow a + (a + 2d) + (a + 4d) + \ldots + (a + 98d) = 50$$

$$\Rightarrow 50a + 2d(1 + 2 + \ldots + 49) = 50$$

$$\Rightarrow 50a + \frac{2d(50)(49)}{2} = 50$$

$$\Rightarrow a + 49d = 1$$

$$= -a_1 - a_3 + a_5 + a_7 - a_9 - a_{11} + \ldots + a_{93} + a_{95} - a_{97} - a_{99}$$

(26 negative terms and 24 positive)

$$\Rightarrow |-a_1 - a_{99}| = |-2a - 98d| \Rightarrow |-2| \Rightarrow 2$$

52. 1

Sol. Given $a + b + c = 1$ .....(1)

$9a + 3b + c = 7$ .....(2)

$18 < 25a + 5b + c < 22$ .....(3)

$\Rightarrow$ From above (1), (2) and (3) $4 < 7a - b - c < 8$

$4 < 7a - b + a - 1 < 8$

$5 < 8a < 9$

$a = 1, b = -1, c = 1$

$a = 1, b = -1, c = 1$

For question (2) $h(x) = \ln(x^2 - x + 1) - x$

53. 4

Sol. Let $z_1 = z_1 e^{i\theta_1}$ and

$$z_2 = r_2 e^{i\theta_2} \Rightarrow \frac{z_1\bar{z}_2 + \bar{z}_1 z_2 + z_1 z_2 + \bar{z}_1\bar{z}_2}{|z_1 z_2|} = 2\cos(\theta_1 + \theta_2) + 2\cos(\theta_1 + \theta_2)$$

54. 3

Sol. $$\frac{a+b+c}{b-a} = \frac{a+b-a+a+c}{b-a}$$

$$= 1 + \frac{2a+c}{b-a}$$

Now: $f(x) = ax^2 + bx + c$

Given $f(x) \geq 0 \ \forall x \ f(-2) \geq 0$

$4a - 2b + c \geq 0$

$2a + c \geq 2(b-a)$

$$\frac{2a+c}{b-a} \geq 2$$

$$\therefore \frac{a+b+c}{b-a} \geq 3$$